Physics-Based Fault Classification in Capillary-Driven Paper Microfluidics

Master document: theory, base paper, sensor, system design, RTL, verification, results and claims

BioMEMS course project · IIIT Allahabad · B.Tech ECE, Semester 7
Group __ · Members: _____
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How to read this document

This is the single source of truth for the project. It is written so that a teammate, or someone new to the project, can follow every step from the physics to the hardware. Suggested routes:

Reader	Read in this order
New to the topic	1 Executive summary → 3 Theory → 4 System concept → 6 Identifiability → 9 Results → 13 Claims
Physics / TA meeting	3 Theory → 5 Calibration → 6 Identifiability (proofs) → 7 Detector → 13 Claims
Hardware / verification	4 System concept → 7 Detector → 11 Architecture and RTL → 12 Verification
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Conventions: L is the distance from the inlet (mm), t the time (s), K the Washburn coefficient (mm^2/s), E_k the expected time for segment k, Δt_k the measured segment time, and $\rho_k = \Delta t_k / E_k$ the segment ratio. A ~ B means A is proportional to B. One tick is 1 ms. Values marked *assumed* have no measured source; all others cite a reference or the computation that produced them.

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1 Executive summary

Problem. Paper rapid diagnostic tests move liquid by capillary wicking alone. When filling goes wrong (too little sample, a defect in the paper, an unusual sample), the test can still produce a result that looks valid. Existing checks reduce the flow to one number: did liquid arrive before a timeout, or how long did one flow take.

Idea. Time the liquid front at four electrodes along a lateral strip. A calibrated capillary-flow model (Lucas–Washburn / Darcy) predicts those times for a healthy strip. The *pattern* of deviations then identifies the kind of fault and where it is, within the limits the physics allows.

What was built.

- A physics model of the strip, calibrated to published wicking data (K = 9.42 mm²/s for nitrocellulose FF170HP).
- A richer generator model (series Darcy resistance, finite sample, fluid changes, strip variability, sensor faults) that produced 9,900 seeded test scenarios.
- A proof-backed identifiability analysis: which faults the timings can and cannot separate.
- A model-based detector/classifier, two baselines, and an ablation, all scored on held-out scenarios.
- A synthesizable SystemVerilog implementation, bit-exact with an independent Python reference on all 9,900 scenarios, verified with assertions, coverage, boundary cases and mutation testing, and implemented on the PYNQ-Z2's Zynq-7020 at 100 MHz.
- A reproduction of the base paper's flow-time algorithm on its published electrode data.

Key numbers

Result	Value	Where
Calibrated K (nitrocellulose, water)	9.42 mm²/s (cross-checks 9.73, 9.41)	§5
Faults detected, this method / fixed timeout / single sensor	88.9 % / 46.8 % / 55.9 %	§9
False alarms on healthy strips (this method)	7.1 % (95 % CI 5.4%–9.1%)	§9
Exact class + location (identifiable classes)	74.5%	§9
RTL vs reference model	9,900 / 9,900 bit-exact; 52 / 52 boundary cases	§12
FPGA cost (xc7z020)	535 LUT, 674 FF, 3 DSP, 0 BRAM; WNS +3.72 ns at 100 MHz	§11
Base-paper flow times reproduced	29.2 / 19.5 / 40.6 s, all inside the paper's IQRs	§10

> **Main limitation**
> The four-electrode results come from a physics model, not from a measured strip. The real data available (the base paper's) has a single electrode, so it validates timing, not classification. See §13.

2 Base paper and its sensor

Reference [1]. L. Bezinge, A. J. deMello, C.-J. Shih, D. A. Richards, "Quantitative reagent monitoring in paper-based electrochemical rapid diagnostic tests", *Lab on a Chip* 24(15), 3651–3657 (2024), doi:10.1039/D4LC00390J. Open access, CC BY-NC 3.0. ETH Zürich (deMello and Richards groups).

2.1 What the base paper does

Graphenized electrodes are embedded in a paper-based electrochemical rapid test. The same electrodes do two jobs:

- **Flow tracking.** Short voltage pulses are applied continuously. When liquid flows over the electrode, the capacitive discharge current changes, which reveals when each injected sample or reagent flows and when it stops.
- **Quantification.** After the last reagent, the reader switches to square-wave voltammetry to read an electrochemical immunoassay (anti-SARS-CoV-2 IgG in serum).

The authors use the flow times to guide the user and to flag incorrect volumes or abnormal viscosities.

2.2 The sensor: how the device is made

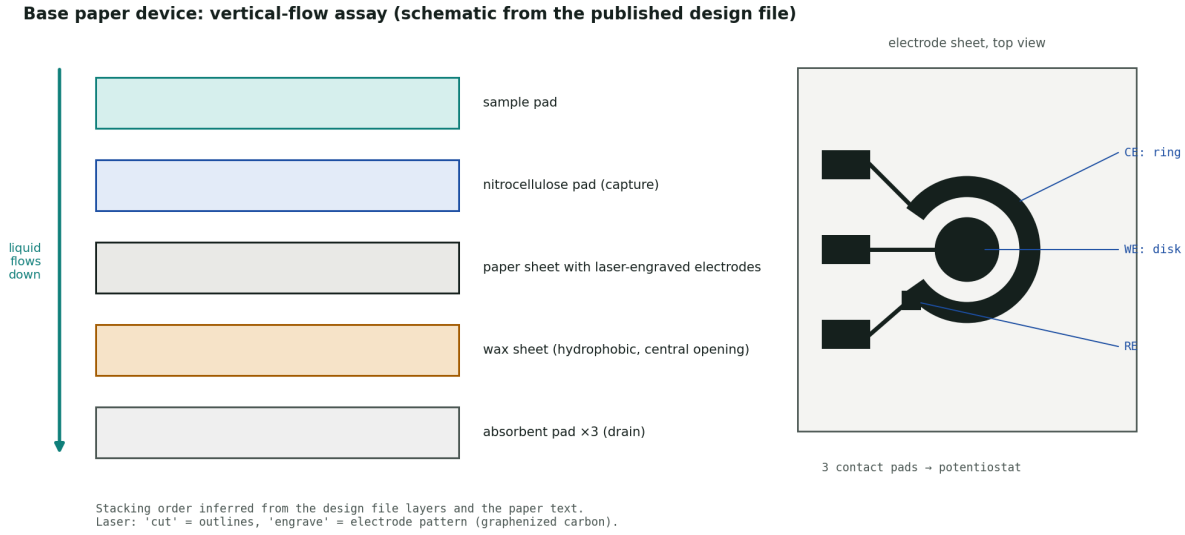


Figure 2.1. Base-paper device, drawn from the published design file (electrofluidic_design_file.pdf). Layer names and the electrode pattern are from the file; the stacking order is inferred from the file and the paper text.

- **Format:** a vertical-flow assay (VFA). Liquid is applied on top and flows down through stacked layers into absorbent pads, which act as a drain.
- **Layers in the design file:** a wax sheet with a central opening and four alignment holes; a paper sheet carrying the electrode pattern; a nitrocellulose pad (the capture membrane of the immunoassay); a sample pad; and three absorbent pads.
- **Fabrication:** the design file is a laser-processing drawing. Its legend marks outlines as *cut* and the electrode pattern as *engrave*. The paper calls the electrodes *graphenized*: the engraved cellulose becomes conductive carbon.
- **Electrode cell:** one three-electrode cell. A disk working electrode (WE) is surrounded by an open ring counter electrode (CE), with a reference electrode (RE); three traces run to contact pads for the potentiostat.

- **Instrument:** a PalmSens potentiostat driven by MethodSCRIPT files (PulseMethod.txt, SWVMethod.txt) through the authors' Python script.

2.3 Measurement principle

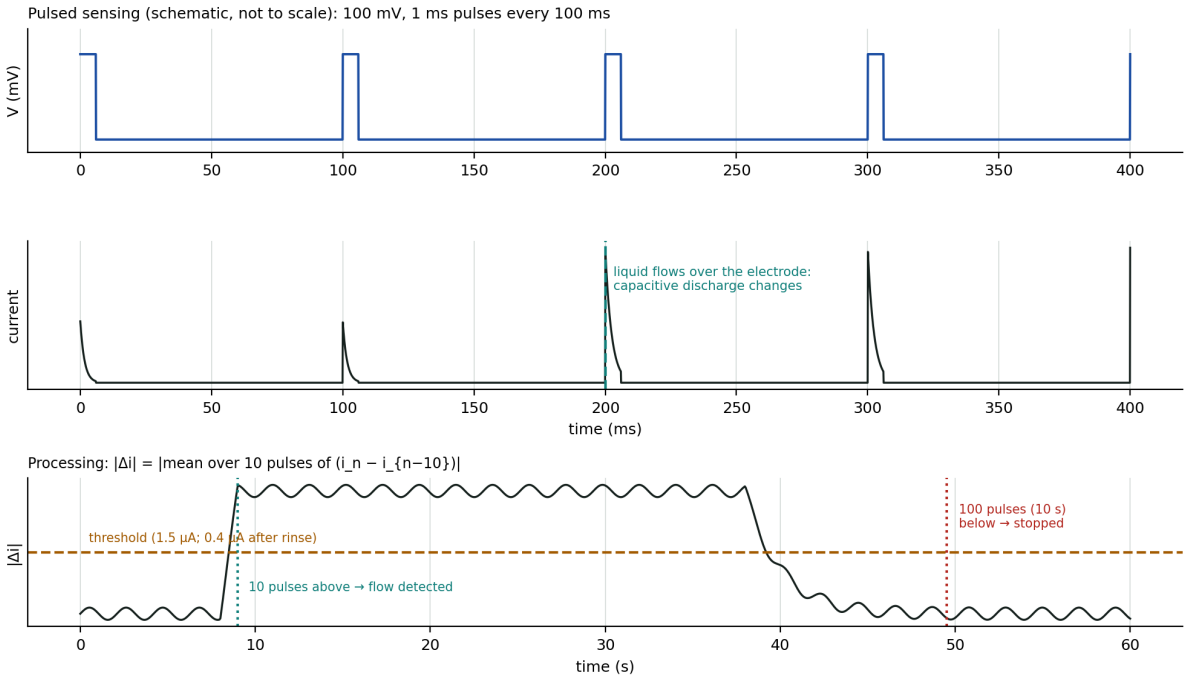


Figure 2.2. Pulsed sensing and processing (schematic, not to scale). Parameters are from PulseMethod.txt and plotResults.m.

Setting	Value	Source
DC potential, pulse height	0 V, 100 mV	PulseMethod.txt (meas_loop_pad)
Pulse length, period	1 ms, every 100 ms (10 samples/s)	PulseMethod.txt; dt = 0.1 s in plotResults.m
Record length	175.1 s	PulseMethod.txt
Current range	auto-ranging 59 nA – 2.95 mA	PulseMethod.txt
Signal	|Δi(n)| = |mean over 10 pulses of (i(n) − i(n−10))|: 1 s difference, 1 s average	plotResults.m; paper text
Thresholds	1.5 µA before the rinse, 0.4 µA after; set as mean + 10 SD of a static liquid	plotResults.m; paper text
Flow detected	10 consecutive samples above threshold	plotResults.m; paper: 'ten-hit minimum'
Flow stopped	100 consecutive samples below (10 s)	plotResults.m; paper: 'ten second time delay'

2.4 Results reported by the base paper

- Electrode flow times were 1.8 s longer on average than visual absorption times (IQR −0.1 to 2.5 s, Fig. S4).
- Flow time scales with added volume: 0.19 ± 0.02 s/µL; 0.36 ± 0.04 s/µL for running buffer (R^2 = 0.91). Dead volume ≈ 20 µL (25 µL did not reach the electrode).
- Albumin at 50 and 200 g/L shortened flow times by factors of 1.5 and 2.5.
- Clinical runs (median, IQR): serum 100 µL 30 s (29–39), labelling 100 µL 22 s (17–32), rinse 150 µL 31 s (28–41).

2.5 The published dataset

The data-availability statement points to Mendeley Data, doi:10.17632/nd9cyw62rk.1. The user supplied these files; they were copied unmodified to `data/bezinge2024/raw/` with SHA-256 checksums. The dataset licence was not checked (user decision), so the files are used non-commercially with attribution and are not redistributed.

File	Content	Used here for
data_P3.csv	Index 0: flow-monitoring current, 1,751 samples at 0.1 s. Indices 1–22: square-wave voltammograms	Real-trace check (§10)
plotResults.m	Flow tracking state machine and SWV peak analysis (MATLAB)	Ported line by line to Python
PulseMethod.txt, SWVMethod.txt	Potentiostat methods	Sampling interval, pulse settings
plot_swv_bipot.py	Instrument control script (needs the potentiostat)	Reference only, not run
electrofluidic_design_file.pdf	Laser cut/engrave drawing of the device layers	Figure 2.1

3 Theory: why liquid moves through paper

3.1 Surface tension and wetting

Molecules at a liquid surface have neighbours on one side only and are pulled inward, so creating surface costs energy. That energy per unit area is the **surface tension** γ (water: 0.072 N/m). Where the liquid meets a solid it forms a **contact angle** θ; θ ≈ 0° means the liquid spreads (wets), and θ > 90° means it beads (as on wax, which is why wax lines form channel walls).

A wetting liquid in a narrow pore of radius r forms a curved meniscus, which carries a pressure jump, the **Laplace pressure**:

> $\Delta P = 2\,\gamma \cos\theta / r$
>
> *Example: water in a 10 µm pore, ΔP ≈ 14 kPa. For the nitrocellulose membrane used here, Kasetsirikul et al. estimate an effective capillary pressure P_c ≈ 2.3 kPa [3].*

3.2 Viscous flow and hydraulic resistance

In a thin tube the flow is laminar, and the mean speed follows Hagen–Poiseuille, where L is the length of liquid in the tube:

> $\bar{v} = r^2 \Delta P / (8\,\mu\,L)$

Viscosity µ (water 0.89 mPa·s, plasma ≈ 1.3, whole blood ≈ 3.5) resists the flow, and the resistance grows in proportion to the wetted length. In the electrical analogy, pressure is voltage, flow rate is current, and each wetted piece of paper is a resistor in series.

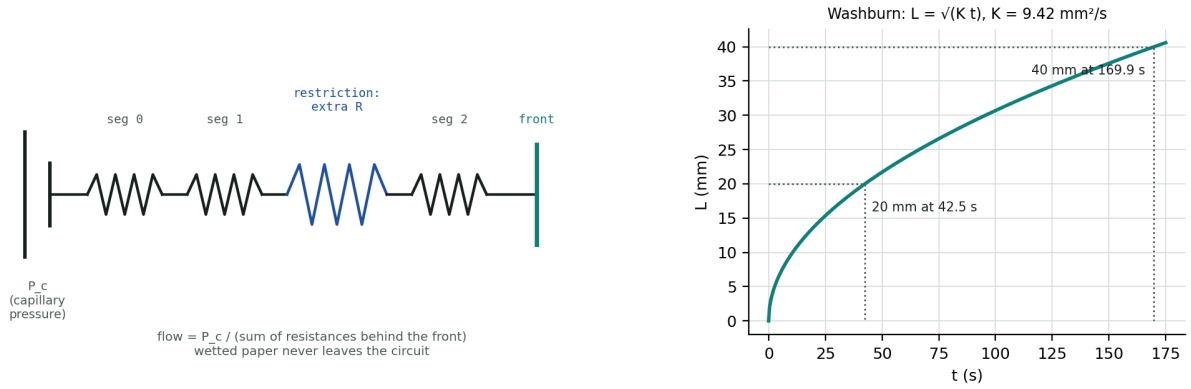


Figure 3.1. Left: electrical analogy; a restriction is an extra series resistor that stays in the circuit. Right: the resulting Washburn curve for K = 9.42 mm²/s (computed).

3.3 Lucas–Washburn law (derivation)

Substitute the capillary pressure into Poiseuille, with $\bar{v} = dL/dt$:

> $dL/dt = (r^2/(8\mu L)) \cdot (2\gamma \cos\theta / r) = r\,\gamma \cos\theta / (4\,\mu\,L)$

Multiplying by L leaves a right-hand side that is constant in time:

> $L \cdot dL/dt = r\,\gamma \cos\theta / (4\mu) \rightarrow d(L^2)/dt = r\,\gamma \cos\theta / (2\mu) \equiv K \rightarrow L^2 = K\,t$
>
> *Lucas (1918), Washburn (1921) [2]. Distance grows as √t: twice the distance takes four times as long. Front speed: dL/dt = K/(2L).*

With K = 9.42 mm²/s, the front reaches 20 mm at 42.5 s and 40 mm at 169.9 s. It moves at 2.4 mm/s near the inlet and 0.12 mm/s at 40 mm.

3.4 Paper as a porous medium: Darcy flow through segments

Paper is a fibre network with a distribution of pore sizes and tortuous paths. The proper description is Darcy's law: flux ~ (permeability / viscosity) × pressure gradient. For a strip of relative permeability k(x) and relative width w(x), the flux Q is the same at every cross-section of the wetted region (quasi-steady, incompressible), so:

> $Q \sim P_c / R(L)$, $R(L) = \int_0^L dx / (k(x)\, w(x))$

The front advances by filling the pores at the front, $\phi \cdot h \cdot W \cdot w(L) \cdot dL/dt = Q$. This gives the generator equation:

> $dL/dt = (K/2) / (w(L) \cdot R(L))$, $K = 2\, k_0\, P_c / (\mu\, \phi)$
>
> *With k = w = 1, R(L) = L and the equation reduces exactly to L·dL/dt = K/2, i.e. Washburn. Integrating gives t(L) = (2/K) ∫ w(x) R(x) dx, which the generator evaluates numerically.*

Because K lumps capillary pressure, permeability, porosity and viscosity together, it is **fitted** to measured wicking rather than computed from a pore radius (§5).

3.5 Finite sample volume

The paper stores liquid at its absorption capacity C_{abs} = 10.1 µL/cm², measured by weighing [3]. A 4 mm wide strip therefore needs 0.404 µL per mm, or 16.2 µL for 40 mm. When the sample is used up, the front creeps on as liquid redistributes and then stops; the generator models the creep as a decay with τ = 20 s (*assumed*).

3.6 Scaling analysis: which forces matter

No source reports this membrane's pore radius, so it is bracketed by two effective radii computed from measured quantities: $r_W = 2\mu K/\gamma$ = 0.23 µm and $r_P = 2\gamma/P_c$ = 61.4 µm. Front speeds are 0.12–2.35 mm/s. The two radii disagree by a factor of about 264, which is itself the reason K must be fitted.

Number	Definition	Low	High	Conclusion
Bond (pore)	rho g r^2 / gamma	7.4e-09	5.1e-04	<< 1: meniscus shape set by capillarity
Capillary	mu U / gamma	1.5e-06	2.9e-05	<< 1: quasi-static capillary pressure
Reynolds (pore)	rho U r / mu	3.1e-05	1.6e-01	<< 1: Stokes/Darcy flow
Inertial time (Bosanquet)	rho r^2 / (8 mu) [s]	7.6e-09	5.3e-04	<< 1 ms tick: inertia invisible to the detector
Strip-scale gravity	L / h_eq, h_eq = P_c/(rho g)	8.4e-03	1.7e-01	irrelevant for a horizontal strip; up to 17 % slowdown if the strip were vertical

Every ratio is well below 1 except the strip-scale gravity term, which matters only for a vertical strip. Surface tension and viscosity govern the flow; gravity and inertia can be neglected. Inertia lasts at most ~0.5 ms, less than one tick.

4 System concept and comparison with the base paper

4.1 The lateral four-electrode strip

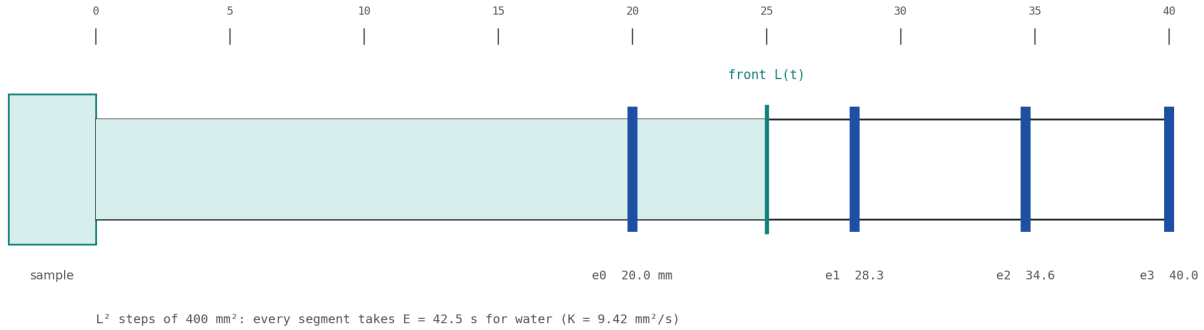


Figure 4.1. Strip layout, to scale: four electrodes at L = 20.0, 28.3, 34.6 and 40.0 mm.

Why these positions. Washburn gives $E_k = (L_k^2 - L_{k-1}^2)/K$. Placing the electrodes where L^2 steps evenly (400, 800, 1200, 1600 mm²) makes every segment take the same expected time, E = 400/9.42 = 42.5 s for water, so each segment has the same timing resolution.

Sensor interface ("interface A"). Each electrode is assumed to drive a comparator whose digital output goes high when the electrode is wetted, for example a base-paper-style pulsed electrode with a threshold detector. The detector receives four asynchronous digital lines. No physical strip was built in this project.

4.2 Base paper versus this project

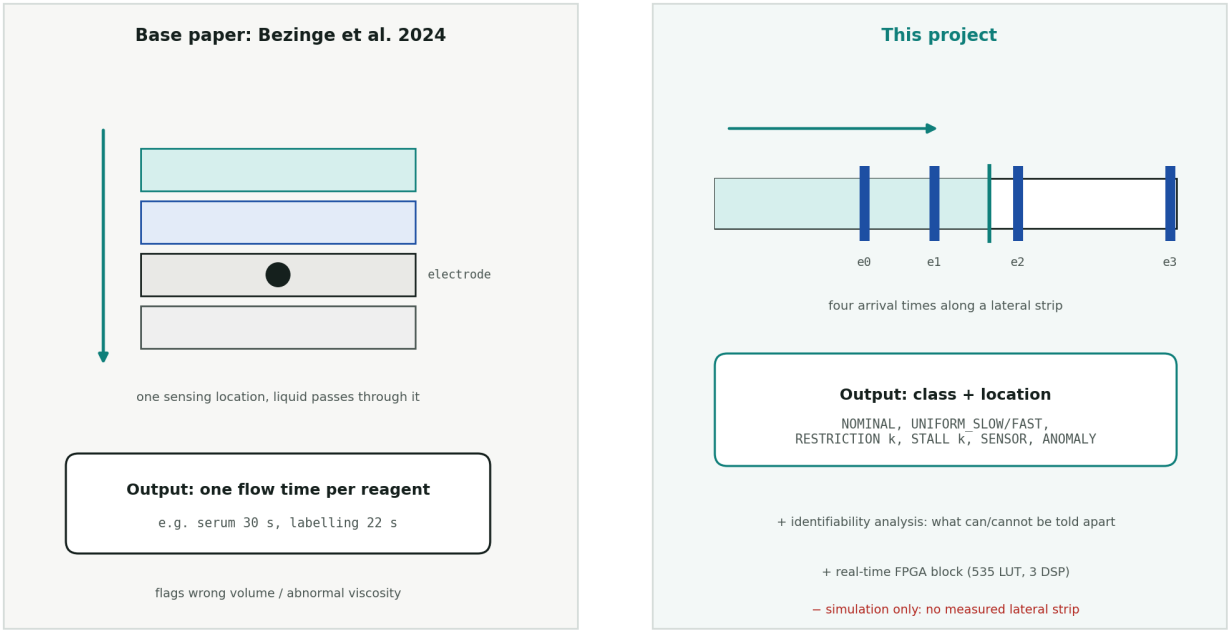


Figure 4.2. Visual comparison: one sensing location in a vertical-flow device against four points along a lateral strip.

Aspect	Base paper [1]	This project
Geometry	Vertical flow through stacked layers	Lateral strip, 40 mm
Sensing points	1 electrode cell	4 electrodes (assumed comparators)
Measured quantity	Flow start/stop per injected reagent	Front arrival time at each electrode

Aspect	Base paper [1]	This project
Output	Flow times; flags wrong volume / viscosity	Class + location: NOMINAL, UNIFORM_SLOW/FAST, RESTRICTION k, STALL k, SENSOR, ANOMALY
Physics model	Empirical thresholds	Calibrated Washburn/Darcy model with an online K estimate
Identifiability analysis	Not addressed	Proofs and computed thresholds (§6)
Processing	Python/MATLAB on a PC	Synthesizable RTL, 535 LUT, verified bit-exact
Validation	Real devices, clinical serum samples	Simulation; real data only for timing (§10)
Quantification (immunoassay)	Yes (square-wave voltammetry)	Out of scope

> **Where the two differ in strength**
> The base paper is stronger on real-world evidence: real devices and clinical samples. This project adds spatial information (where a fault is), an explicit analysis of what can and cannot be identified, and a hardware implementation, but it is supported only by simulation.

4.3 End-to-end pipeline of the project

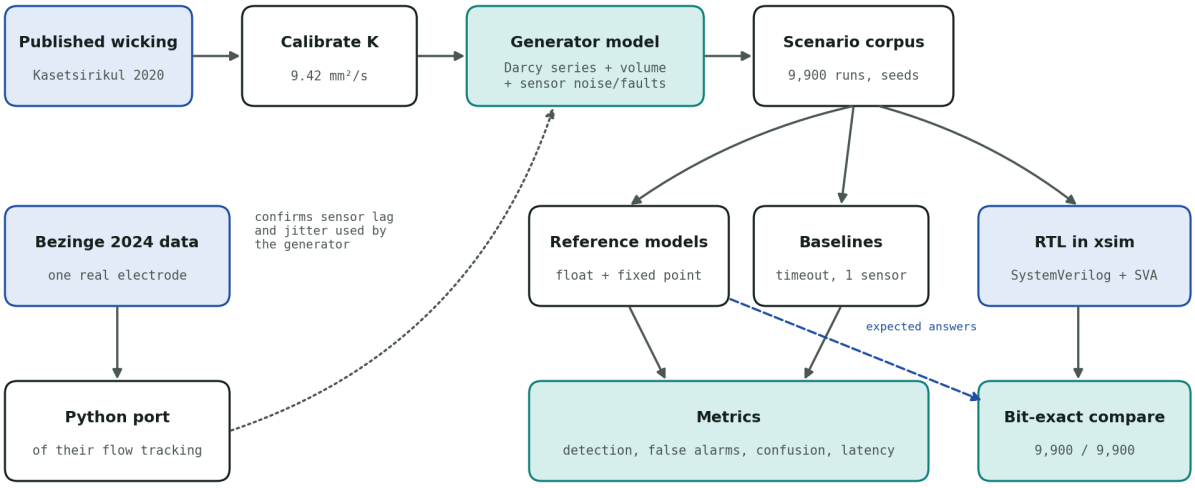


Figure 4.3. How the parts fit together. The dashed blue arrow: the reference model supplies the expected answer for every RTL run.

5 Calibration and generator model

5.1 Data source

Publisher pages for most wicking studies (PMC, ACS, RSC) block automated access. The open-access paper by Kasetsirikul, Shiddiky and Nguyen (*Chemosensors* 8, 65, 2020, CC BY) [3] reports wicking of dyed water in nitrocellulose FF170HP strips 2, 4 and 6 mm wide, n = 3 each, as a vertical capillary rise. 33 points were read from its Fig. 7a (±0.5 mm) and stored with provenance in `data/literature/kasetsirikul2020/`.

5.2 Fits

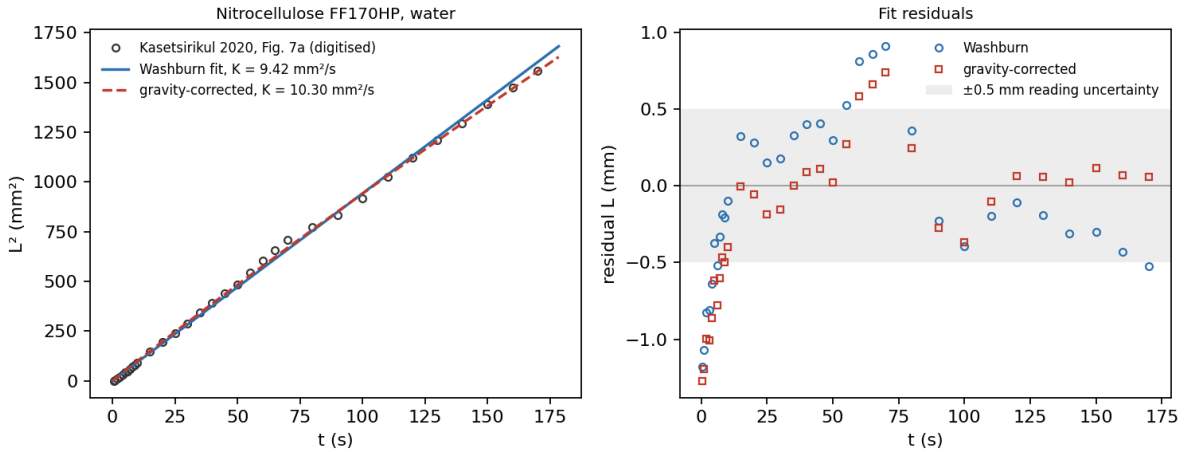


Figure 5.1. L^2 against t with the Washburn fit and the gravity-corrected fit (left), and residuals (right).

Fit / check	K (mm²/s)	Note
Washburn through the origin, L = √(Kt)	9.42	adopted; rms residual 0.53 mm
Paper's own fit p^2 (Table 6)	9.73	+3 %
Product rating, 4 cm in 170 s	9.41	−0.1 %
Gravity-corrected, $h_{eq} = P_c/\rho g$ = 239 mm	10.3	+9 %: carried as a systematic uncertainty
Power-law exponent n (Washburn 0.5)	0.546	early-time deviation of about 1 mm in the first 10 s
CV of K across strip widths	3.2 %	strip-to-strip variability

5.3 Generator model

- **Flow:** the Darcy series equation of §3.4, integrated on a 1 µm grid. A restriction is a segment with k < 1; a narrowed neck is a segment with w < 1.
- **Fluid:** K scales with γ cos θ / µ relative to water.
- **Variability:** K multiplied by a log-normal factor with CV 0, 3.2 % or 10 %.
- **Finite volume:** the sample runs out when the wetted volume equals it; creep follows with τ = 20 s.
- **Sensor:** a constant detection lag of 1.8 s, Gaussian jitter σ in {0, 0.1, 0.5, 1, 2} s, and missed, spurious (early random trigger) or stuck (high before arm) electrodes.

Seven automated checks guard the generator (`run.py phase2`). A uniform strip reproduces $L^2 = Kt$ to 10^{-4}. A k = 1 segment changes nothing. Viscosity scales all arrivals equally. A restriction leaves upstream arrivals unchanged and delays downstream ones. An 8 µL sample stops at 24.1 mm (the volume limit is 19.8 mm, the rest is creep). A 12 µL sample never reaches 40 mm. A narrow neck slows the front downstream.

5.4 Parameters and their sources

Parameter	Value	Unit	Provenance
K_NOM	9.42	mm^2/s	repo: physics/calibrate.py Washburn fit to KAS20 Fig. 7a (digitised); cross-checks KAS20 Table 6 p^2 = 9.73, FF170HP rating 40^2/170 = 9.41; gravity-corrected fit +9 % (systematic)
K_CV	0.032	-	repo: CV of K = p^2 over the 2/4/6 mm strips, KAS20 Table 6
K_CV_HIGH	0.1	-	PAT20: max deviation < 10 % across five runs (Whatman 4), used as upper bound
P_C	2345.0	Pa	KAS20 Table 8 (nitrocellulose, model-derived estimate)
C_ABS	0.101	uL/mm^2	KAS20 Sec. 4.3: 10.1 uL/cm^2, direct weighing, non-laminated NC
EXPONENT	0.546	-	repo: power-law fit L ~ t^n to KAS20 Fig. 7a (t >= 1 s); MIR26 reports 0.39-0.46 for radial imbibition in filter paper over 60 min
GAMMA	0.072	N/m	PAT20 Table (water)
MU	0.00089	Pa s	PAT20 / MIR26 (water); KAS20 uses 8.94e-4
RHO	1000.0	kg/m^3	KAS20 Table 2
THETA	0.0	deg	assumed: fully wetting (PAT20 makes the same assumption); enters only via K
L_MAX	40.0	mm	assumed: equals the 4 cm FF170HP rating length and the KAS20 strip
WIDTH	4.0	mm	assumed: middle KAS20 width (wicking speed independent of width, KAS20 Sec. 4.4)
N_ELEC	4	-	assumed: handoff default; electrodes spaced uniformly in L^2
V_SAMPLE	25.0	uL	assumed: typical RDT sample volume; strip needs C_ABS*WIDTH*L_MAX = 16.2 uL
CREEP_TAU	20.0	s	assumed: decay time of front creep after the source empties (KAS20 Fig. 2 shows the front keeps moving after reservoir removal; magnitude not reported)
TICK_S	0.001	s	design choice: fill takes ~10^2 s, so 1 ms is ample
LAG_S	1.8	s	BEZ24 main text (verified in Phase 6): electrode flow times average 1.8 s longer than visual absorption (IQR -0.1 to 2.5 s, Fig. S4); applied here as a constant arrival-detection lag (an interpretation: the paper's figure is a duration bias)
JITTER_S	(0.0, 0.1, 0.5, 1.0, 2.0)	s	sweep of arrival-detection jitter (1 sigma); BEZ24's IQR implies ~1.4 s per detected event (realdata/compare.py), inside the swept range
MU_PLASMA	0.0013	Pa s	assumed: typical plasma/serum viscosity 1.1-1.3 mPa s (textbook value, verify)
MU_BLOOD	0.0035	Pa s	assumed: whole blood at high shear 3-4 mPa s (textbook value, verify)
MU_WATER37	0.00069	Pa s	assumed: water at 37 C (standard tables)

6 Identifiability: what the timings can and cannot reveal

The detector sees only arrival times, and therefore only the segment ratios $\rho_k = \Delta t_k / E_k$. This chapter states what can be inferred from them in principle, before any algorithm is involved.

6.1 Proposition 1: the fluid property behind a K change is not identifiable

Claim. Two fluids with the same value of $\gamma \cos\theta / \mu$ produce identical arrival times on any strip.

Proof. In the generator equation, fluid properties enter only through $K \sim \gamma \cos\theta / \mu$; the geometry terms w(L) and R(L) do not depend on the fluid. Equal K means an identical ODE, hence identical L(t) and arrival times. QED

Demonstration. Doubling viscosity, halving surface tension, and raising θ from 0° to 60° (cos 60° = 0.5) each halve K. Through the generator, the sensor model and the fixed-point detector, all three give the ticks 84926, 169852, 254778, 339703 and the verdict UNIFORM_SLOW.

6.2 Proposition 2: signature of a localized restriction

Let a restriction add ΔL of equivalent length, $\Delta L = (1/k - 1)\cdot w$ for relative permeability k over width w. For x beyond the restriction, $R(x) = x + \Delta L$, so $t(L) = (L^2 + 2\Delta L \cdot L + c)/K$, where c is a constant. A segment that lies entirely downstream of the restriction then has:

> $$\rho_k = 1 + 2\Delta L \,/\, (L_k + L_{k-1})$$
>
> *Segments upstream are unaffected (ρ = 1); the segment containing the restriction is partly affected.*

Consequences: (i) the first deviating segment locates the restriction; (ii) the same ΔL produces a smaller signature further downstream, because $L_k + L_{k-1}$ grows; (iii) a restriction below some ΔL is hidden by the tolerance.

6.3 Proposition 3: restriction upstream of the first electrode

If the restriction lies before electrode 0, every segment is downstream of it, so all ρ_k exceed 1 and **decrease** along the strip by the formula above. A large one produces this non-Washburn pattern, which the detector reports as ANOMALY. A small one gives nearly flat ratios, indistinguishable from a slower fluid (Proposition 1).

6.4 Computed thresholds

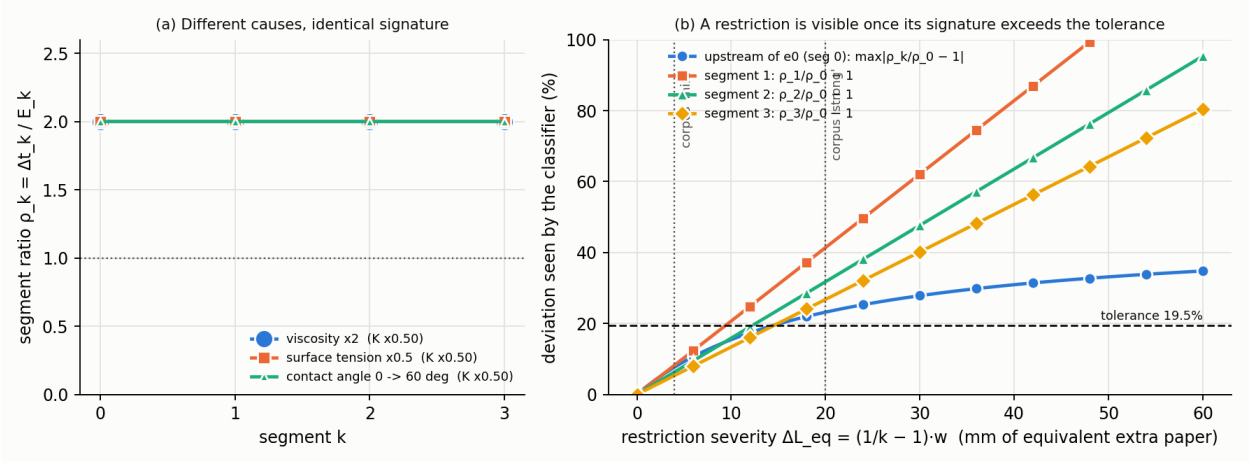


Figure 6.1. (a) Three different causes, identical signature. (b) Deviation seen by the classifier against restriction severity ΔL_{eq}, noise-free, with the 19.5 % tolerance.

Noise-free, a restriction is reported in the right segment once ΔL_{eq} exceeds 10 mm (segment 1), 13 mm (segment 2) or 15 mm (segment 3). An upstream restriction becomes ANOMALY above about 15 mm, and its deviation saturates near 35 %. The corpus's 'mild' restrictions are 4 mm (below every threshold); its 'strong' ones are 20 mm (above all).

6.5 Identifiability table (with test-split evidence)

Fault class	Signature across 4 electrodes	Identifiable?	Evidence
Fluid-property change: viscosity, surface tension, contact angle	All segment ratios equal, ρ_k = ρ_0 ≠ 1: Washburn with a different K	**Class: yes. Which property: no**	Test split: slow fluids flagged UNIFORM_SLOW 100%; warm water (K +29 %) flagged UNIFORM_FAST 87%. Viscosity ×2, surface tension ×0.5 and contact angle 0→60° give identical arrival ticks [84926, 169852, 254778, 339703], so no detector can separate them.
Localized restriction in segment j ≥ 1	ρ_0 … ρ_{j−1} nominal; ρ_j steps up, later segments stay raised	**Yes, with location**, above a severity threshold	Noise-free threshold ΔL_eq: segment 1: 10 mm, segment 2: 13 mm, segment 3: 15 mm. Strong (20 mm): detected 94%, correct segment 90%. Mild (4 mm, below threshold): detected 13%, barely above the 7% false-alarm rate on nominal strips. Segment 3 strong is detected least, 81%: 20 mm sits close to its 15 mm threshold.
Restriction upstream of electrode 0	Adds a constant resistance: $t \sim L^2 + 2 \cdot \Delta L \cdot L$, so ρ_k/ρ_0 decreases along the strip	**Partly.** Weak: no, it looks like a fluid change. Strong: flagged as non-Washburn	Noise-free: indistinguishable from UNIFORM_SLOW below ΔL_eq ≈ 15 mm, ANOMALY above. Test split: ANOMALY 52%, UNIFORM_SLOW 37%, NOMINAL 10%. The deviation saturates near 35 %, so it is never as visible as a downstream restriction.
Insufficient volume / stall	Downstream electrodes never fire within the adaptive window	**Yes, with the gap**	Test split: STALL with the correct gap 100%.
Sensor fault: missed, spurious, stuck	Impossible ordering, a line high before the sample, or an implausibly fast first segment	**Usually.** Electrode index is ambiguous	Test split: flagged as a fault 100%, classified SENSOR 90%, correct electrode 65%. A missed upstream electrode and a spurious downstream one give the same out-of-order pattern. A missed last electrode is physically a stall in the last gap (STALL 100% of those cases).
K change smaller than the strip-to-strip spread	Flat ρ inside the nominal band	**No**, by design	The ±14.8% band absorbs the measured 3.2–10 % batch variability and the +9 % gravity systematic on K, at the cost of fluids within that band of water.

7 Detector algorithm

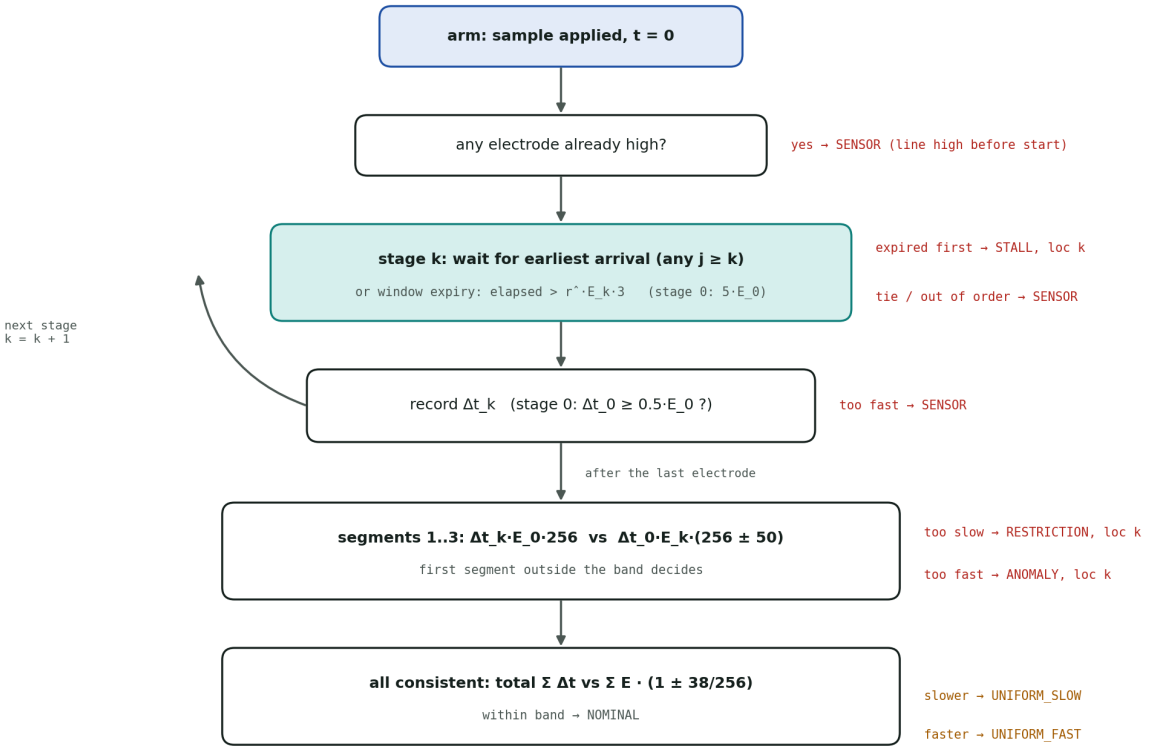


Figure 7.1. Decision flow of the detector (SPEC.md v0.2).

7.1 Rules

- **R0, pre-high:** an electrode line already high when the sample is applied → SENSOR (lowest such index).
- **R1, stage window:** while waiting for electrode k, STALL at k if the elapsed time e since the previous arrival satisfies $e \cdot 256 > E_0 \cdot$FIRST_Q (k = 0) or $e \cdot E_0 \cdot 256 > \Delta t_0 \cdot E_k \cdot (256 + \text{TOL_STALL})$ (k ≥ 1). An arrival that already satisfies this is too late: timeout wins ties.
- **R2, order:** if the earliest arrival among electrodes j ≥ k is not k alone (a tie, or another electrode first) → SENSOR at the offending electrode.
- **R3, accept:** record Δt_k; at k = 0, if $\Delta t_0 \cdot 256 < E_0 \cdot$MIN_FIRST_Q → SENSOR (faster than any real sample).
- **Classify** (after the last electrode): for k = 1..N−1 in order, if $\Delta t_k \cdot E_0 \cdot 256 > \Delta t_0 \cdot E_k \cdot (256 + \text{TOL})$ → RESTRICTION k; if $< \Delta t_0 \cdot E_k \cdot (256 - \text{TOL})$ → ANOMALY k.
- **Uniform band:** otherwise compare the total time $T = \Sigma \Delta t$ with $E_{TOT} = \Sigma E$: above $E_{TOT} \cdot (256 + \text{TOL_ABS})/256$ → UNIFORM_SLOW; below the lower band → UNIFORM_FAST; else NOMINAL.

7.2 Why these choices

- **Online K.** $\hat{r} = \Delta t_0 / E_0$ measures the sample's own speed, so a viscous but healthy sample (blood ≈ 3.9× water) is accepted and later windows scale with it.
- **Adaptive windows.** Each stage's window depends on $\hat{r}$, which is what lets the detector name the gap where the front stopped; a fixed global timeout cannot.
- **Total-time uniform test (v0.2).** A constant 1.8 s detection lag is 4 % of the first segment but only 1 % of the total, so testing the total raised fast-fluid detection from 47 % to 92 %.

- **First-segment floor (v0.2).** No plausible sample wicks more than twice as fast as water (water at 37 °C is 1.29×). This floor removed false STALLs caused by spurious early triggers.

7.3 Exactness of the arithmetic (why hardware can match software exactly)

Claim. Every decision is an exact integer comparison, so any correct implementation, in Python or RTL, gives identical results. **Argument.** The ratio tests are cross-multiplied: there is no division and no rounding, and all tolerances are integers over 256. Inputs are integer ticks. The largest product is $\Delta t_0 \cdot E_k \cdot 768 < 2^{24} \cdot 2^{24} \cdot 2^{10} = 2^{58}$, so a 64-bit comparator is exact. The Python model asserts every product is below 2^{64}. QED

> **Worked example**
> A restriction in segment 2 (ticks in ms): Δt = 42463, 42463, 55960, 65219; all E = 42463. Segment 1: 42463·256 = 10,870,528 lies between 42463·206 and 42463·306, so it is consistent. Segment 2: 55960·256 = 14,325,760 > 42463·306 = 12,993,678, so the verdict is RESTRICTION in segment 2.

7.4 Parameters

Parameter	Value	Meaning	Origin
TOL	50/256 = 19.5%	segment consistency	tuned on calibration seeds
TOL_ABS	38/256 = 14.8%	nominal band on total time	tuned on calibration seeds
TOL_STALL	512/256	window = $\hat{r} \cdot E_k \cdot 3$	design choice
FIRST_Q	1280/256	first segment ≤ $5 \cdot E_0$	design choice (blood 3.9×)
MIN_FIRST_Q	128/256	first segment ≥ $0.5 \cdot E_0$	physics bound
E_k	42463, 42463, 42463, 42463 ticks	expected segment times	L^2 spacing, K = 9.42
Tick / timestamp width	1 ms / 24 bit	4.66 h range	design choice

8 Dataset and experiments

8.1 Simulated scenario corpus

The corpus contains 29 fault scenarios × 5 jitter levels [0.0, 0.1, 0.5, 1.0, 2.0] s × 3 K-variability levels [0.0, 0.032, 0.1] × 20 seeds (100 for nominal strips) = 9,900 runs. Seeds are derived from the cell indices, so the corpus is bit-reproducible. Replicates in the first half form the **calibration** split and the rest the **test** split.

Scenario family	Truth	Physical setting
nominal	NOMINAL 0	healthy strip, water
uniform_water37C	UNIFORM_FAST 0	K × 1.29 (fluid property change)
uniform_plasma	UNIFORM_SLOW 0	K × 0.68 (fluid property change)
uniform_2x_viscosity	UNIFORM_SLOW 0	K × 0.50 (fluid property change)
uniform_whole_blood	UNIFORM_SLOW 0	K × 0.25 (fluid property change)
restriction_seg0_mild	RESTRICTION 0	permeability ×0.33 over 2 mm centred at 10.0 mm (ΔL_{eq} = 4 mm)
restriction_seg0_strong	RESTRICTION 0	permeability ×0.17 over 4 mm centred at 10.0 mm (ΔL_{eq} = 20 mm)
restriction_seg1_mild	RESTRICTION 1	permeability ×0.33 over 2 mm centred at 24.1 mm (ΔL_{eq} = 4 mm)
restriction_seg1_strong	RESTRICTION 1	permeability ×0.17 over 4 mm centred at 24.1 mm (ΔL_{eq} = 20 mm)
restriction_seg2_mild	RESTRICTION 2	permeability ×0.33 over 2 mm centred at 31.5 mm (ΔL_{eq} = 4 mm)
restriction_seg2_strong	RESTRICTION 2	permeability ×0.17 over 4 mm centred at 31.5 mm (ΔL_{eq} = 20 mm)
restriction_seg3_mild	RESTRICTION 3	permeability ×0.33 over 2 mm centred at 37.3 mm (ΔL_{eq} = 4 mm)
restriction_seg3_strong	RESTRICTION 3	permeability ×0.17 over 4 mm centred at 37.3 mm (ΔL_{eq} = 20 mm)
stall_seg0	STALL 0	sample volume 0.10 µL (front stops mid-segment)
stall_seg1	STALL 1	sample volume 8.02 µL (front stops mid-segment)
stall_seg2	STALL 2	sample volume 11.44 µL (front stops mid-segment)
stall_seg3	STALL 3	sample volume 14.02 µL (front stops mid-segment)
sensor_missed_e0	SENSOR 0	electrode 0 missed
sensor_spurious_e0	SENSOR 0	electrode 0 spurious
sensor_stuck_e0	SENSOR 0	electrode 0 stuck
sensor_missed_e1	SENSOR 1	electrode 1 missed
sensor_spurious_e1	SENSOR 1	electrode 1 spurious
sensor_stuck_e1	SENSOR 1	electrode 1 stuck
sensor_missed_e2	SENSOR 2	electrode 2 missed
sensor_spurious_e2	SENSOR 2	electrode 2 spurious
sensor_stuck_e2	SENSOR 2	electrode 2 stuck
sensor_missed_e3	SENSOR 3	electrode 3 missed
sensor_spurious_e3	SENSOR 3	electrode 3 spurious
sensor_stuck_e3	SENSOR 3	electrode 3 stuck

8.2 Methods compared

Method	Decision rule	Tuned parameter
Model-based (this work)	Rules of §7	TOL = 50/256, TOL_ABS = 38/256
Fixed timeout	fault if the last electrode does not fire before T	T = 188.5 s
Single sensor	fault if the last arrival is outside [T_{lo}, T_{hi}]	149.2–195.1 s
Fixed-K windows (ablation)	fault if any ρ_k is outside 1 ± tol (no online K)	tol = 0.19

8.3 Tuning protocol

Every method is tuned on the calibration split to a false-alarm rate of at most 5 % on nominal strips (all noise levels pooled). For the model-based detector, a grid search over (TOL, TOL_ABS) maximizes exact class + location accuracy on calibration faults under that constraint. All reported metrics come from the unseen test split. `run.py phase3` fails if the tolerances in params.py differ from the tuned values.

8.4 Metrics

- **Detection rate:** fraction of faulty test strips flagged as anything other than NOMINAL.
- **False-alarm rate:** fraction of healthy test strips flagged.
- **Exact accuracy:** class and location both correct (upstream restrictions excluded as not identifiable).
- **Time to decision:** tick at which the result is final (the stall decision tick is computed by the reference model).

9 Results

9.1 Detection against baselines (test split, 4,950 runs)

Method	Faults detected	False alarms	Classifies?
model-based	88.9 %	7.1 %	cause + location
fixed timeout	46.8 %	7.3 %	no
single sensor	55.9 %	6.4 %	no
fixed-K windows	89.4 %	6.7 %	no

The model-based method's test false-alarm rate is 53/750 = 7.1% (Wilson 95 % CI 5.4%–9.1%). At equal false-alarm rates, the four-electrode methods detect about twice as many faults as either single-number check. The fixed-K ablation detects equally well, so the gain in *detection* comes from the extra electrodes. The model-based method adds *classification*: exact class and location in 74.5% of cases.

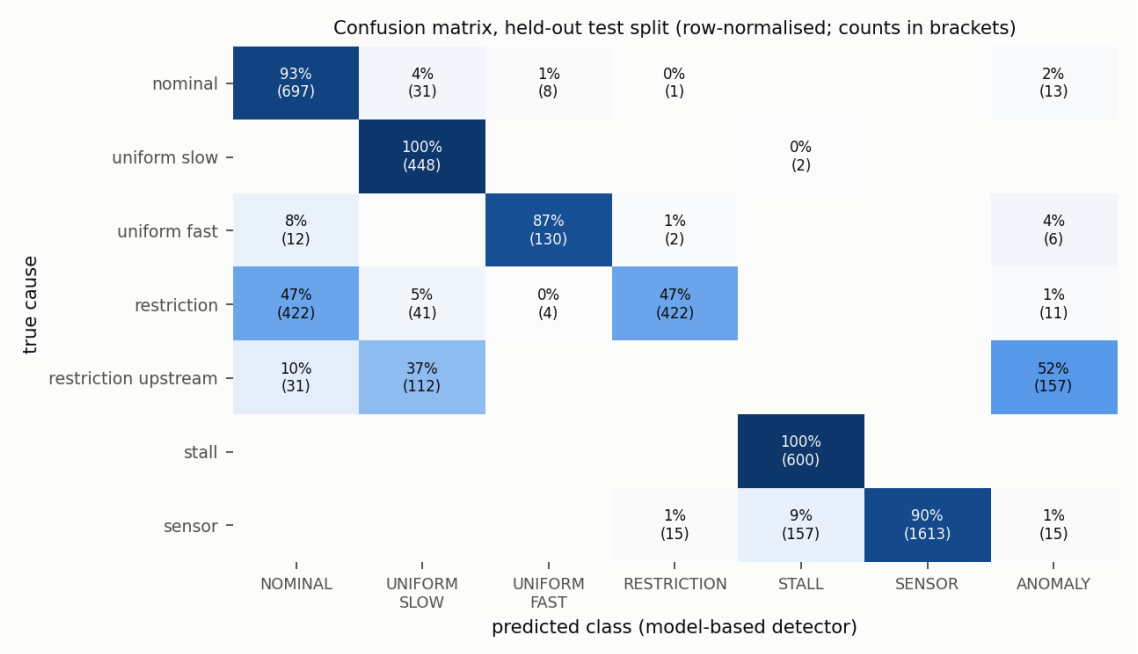


Figure 9.1. Confusion matrix, test split (row-normalized; counts in brackets).

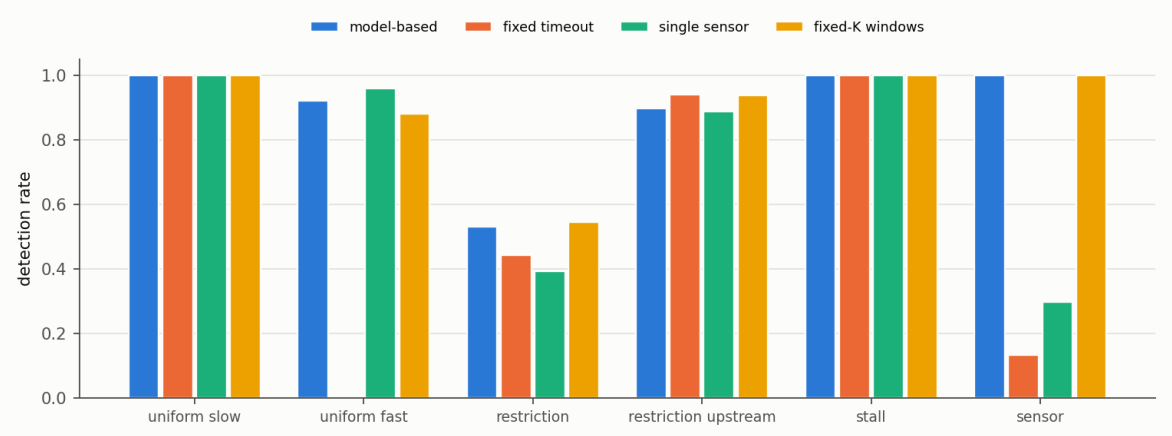


Figure 9.2. Detection rate by true cause for each method.

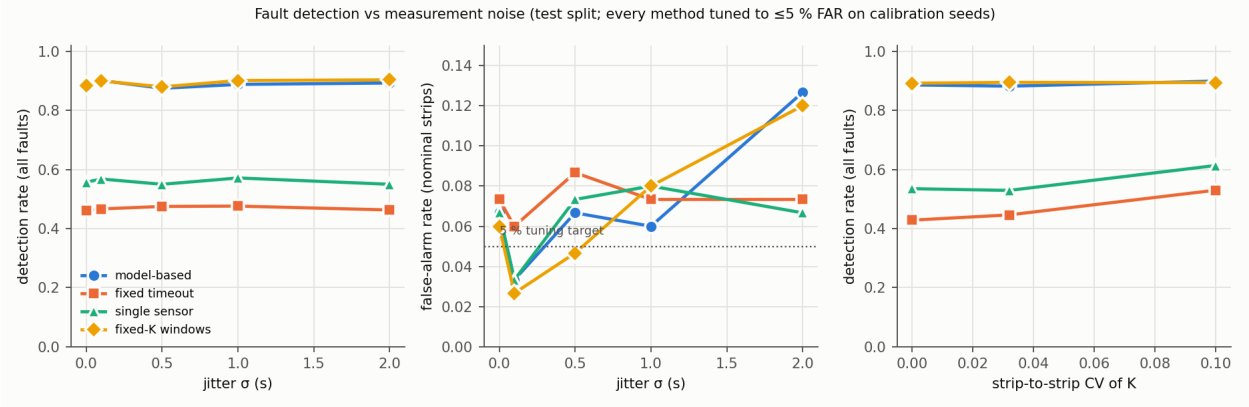


Figure 9.3. Detection and false alarms against jitter and strip variability. Real electrodes sit at about σ = 1.4 s (§10).

9.2 Time to decision (median, seconds)

True cause	model-based	fixed timeout	single sensor	fixed-K
nominal	172	172	172	172
uniform slow	342	188	195	87
uniform fast	133	133	133	68
restriction	186	186	186	173
restriction upstream	244	188	195	83
stall	212	188	195	nan
sensor	31	172	172	29

The model-based method waits for the full run before classifying viscous samples (median 342 s), and it reports sensor faults early (31 s).

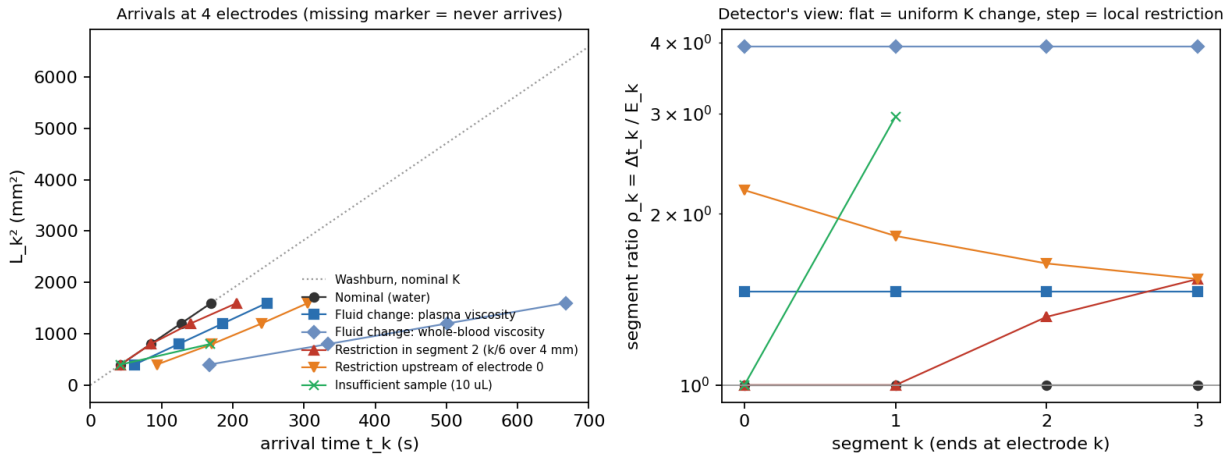


Figure 9.4. Arrival patterns per fault class from the generator (no noise): flat ratios mean a uniform K change; a step means a localized restriction.

10 Real-data validation against the base paper

Method. The flow-tracking section of `plotResults.m` was ported line by line to Python (`realdata/bezinge_flow.py`), keeping the MATLAB's quirks, for example a stop counter that is never reset. Octave was not installed, so instead of a side-by-side run the port is checked against an analytic case.

Self-test proof. For a current step of height h >> threshold at sample s, i(n) − i(n−10) = h for n in [s, s+9]. So |Δi| is a triangle above the threshold for n in [s, s+18]. The 10th consecutive hit is at s+9 (flow detected). The signal falls below the threshold from s+19, so the 100th consecutive low sample is at s+118 (stop). The port returns exactly (s+9, s+118).

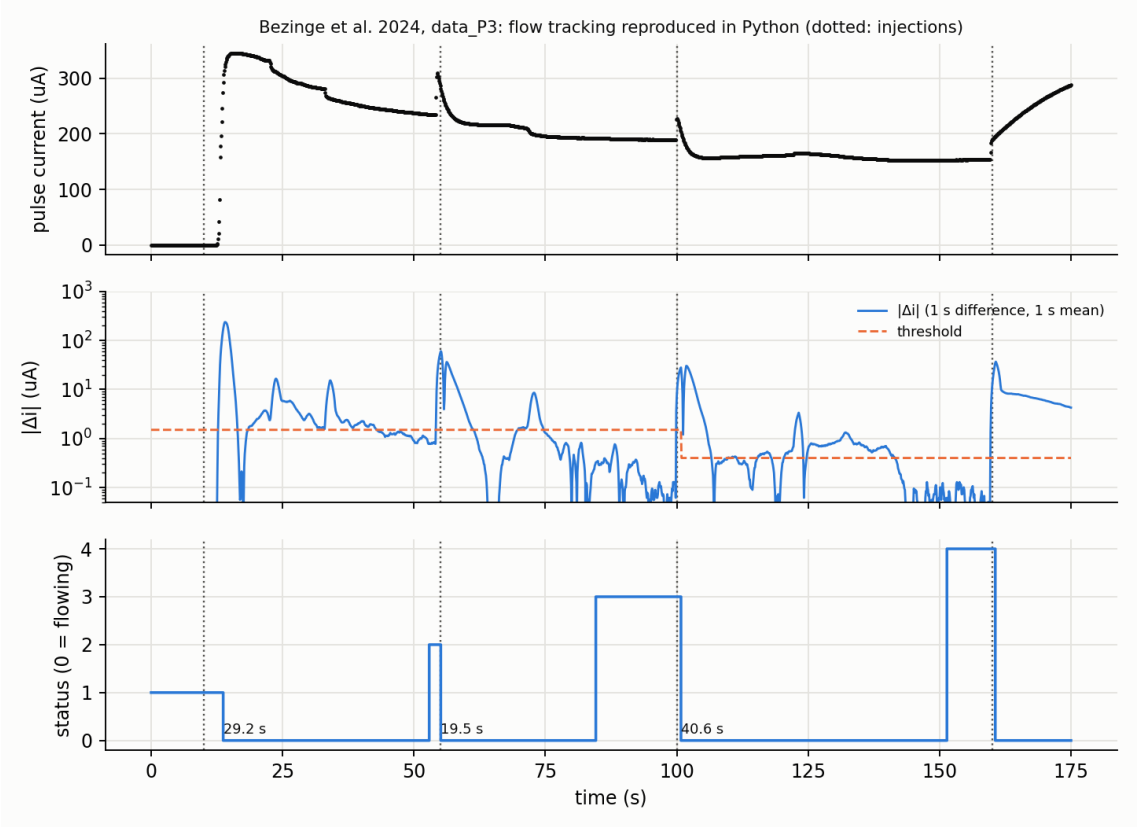


Figure 10.1. Reproduction on data_P3.csv: current, |Δi| with threshold, and the status trace with flow times.

Injection	Step (inferred)	Injected at (s)	Flow detected (s)	Stop declared (s)	Flow time, reproduced (s)	Paper median (IQR) (s)	In IQR
1	serum sample, 100 uL	10	13.7	52.9	29.2	30 (29-39)	yes
2	labelling solution, 100 uL	55	55.1	84.6	19.5	22 (17-32)	yes
3	rinsing buffer, 150 uL	100	100.8	151.4	40.6	31 (28-41)	yes
4	substrate (reader switches to SWV)	160	160.6	-	n/a (no stop)	n/a	-

Flow time = (stop declared) − (flow detected) − 10 s confirmation delay. The step names are inferred from the assay order described in the paper (sample, labelling, rinse, substrate). All three completed flow times fall inside the paper's reported IQRs.

Implied sensor noise. The paper's agreement with visual timing has an IQR of −0.1 to 2.5 s. For a normal distribution IQR = 1.349 σ, so σ ≈ 1.9 s per flow time, or ≈ 1.4 s per detected event if the two events are independent (√2). The 1.8 s bias sets the generator's lag. The 1.4 s jitter lies inside the simulated 0–2 s range.

> **Scope of this validation**
> The base-paper device has a single electrode cell and vertical flow. This check validates event timing and its noise; it cannot test multi-electrode localization or the lateral calibration.

11 System architecture and RTL

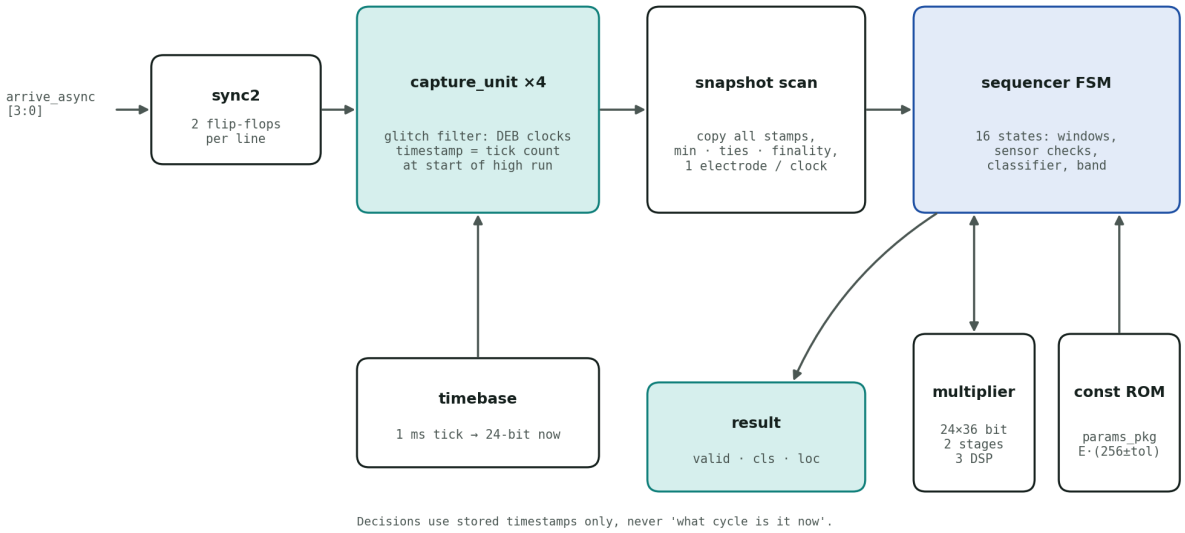


Figure 11.1. RTL block diagram.

11.1 Modules

File	Lines	Role
`rtl/sync2.sv`	12	2-flip-flop synchronizer per electrode line (metastability)
`rtl/capture_unit.sv`	40	glitch filter (DEB clocks) + timestamp at the start of the high run; first rise only
`rtl/mul_u.sv`	18	shared unsigned multiplier 24×36 bit, 2 register stages (DSP48E1)
`rtl/detector_core.sv`	178	timebase, snapshot scan, 16-state sequencer, classifier, uniform band
`rtl/detector_top.sv`	18	sync2 + core; ports clk, rst_n, tick, arm, arrive_async[3:0], valid, cls, loc
`rtl/detector_syn_top.sv`	33	synthesis wrapper: 1 ms tick generator (100,000 clocks), DEB = 1000 (10 µs)
`build/*/params_pkg.sv`	gen.	generated constant ROM: E_k, $E_k \cdot (256 \pm TOL)$, stall constants, bands
`tb/detector_core_checks.svh`	197	assertions + covergroups, compiled in simulation only
`tb/tb_top.sv`	76	file-driven testbench; tick per clock; glitch injection

11.2 Timestamp-only decisions (correctness argument)

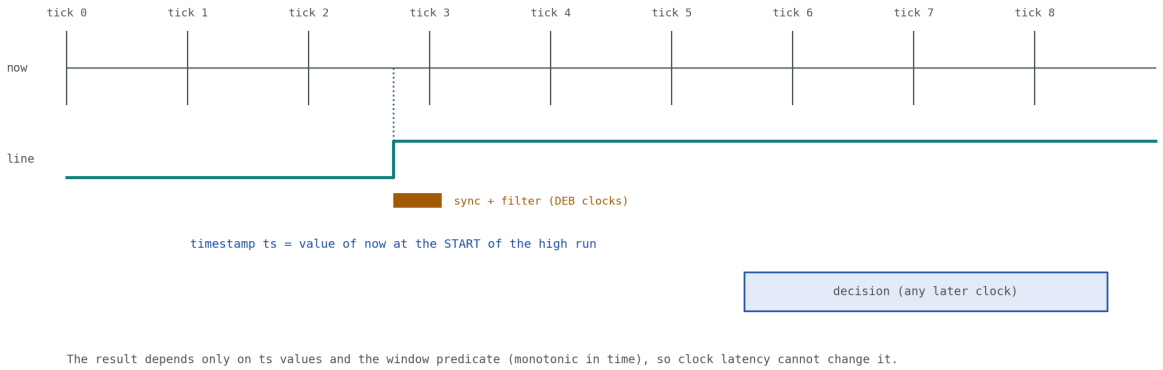


Figure 11.2. The timestamp is taken at the start of the high run; the decision can come later.

Claim. The RTL's result equals the SPEC function of the arrival ticks, whatever the clock-to-tick ratio or multiplier latency. **Argument.** (1) Each capture unit stores ts = now at the first clock of a high run that lasts

DEB clocks, so ts equals the arrival tick if no tick boundary falls within the synchronizer latency (the testbench guarantees this; in silicon a tick is 10^5 clocks). (2) An arrival at tick m is acted on only when no filter is pending and now > m, so no other electrode can still be stamped m: the earliest arrival and its tie set are final. (3) A timeout is declared when the window predicate holds at a sampled now; any later capture has ts ≥ now, and the predicate is monotonic in time, so that capture would also be late. (4) All arithmetic is exact (§7.3). Hence decisions depend only on the ts values. QED

11.3 Sequencer state machine

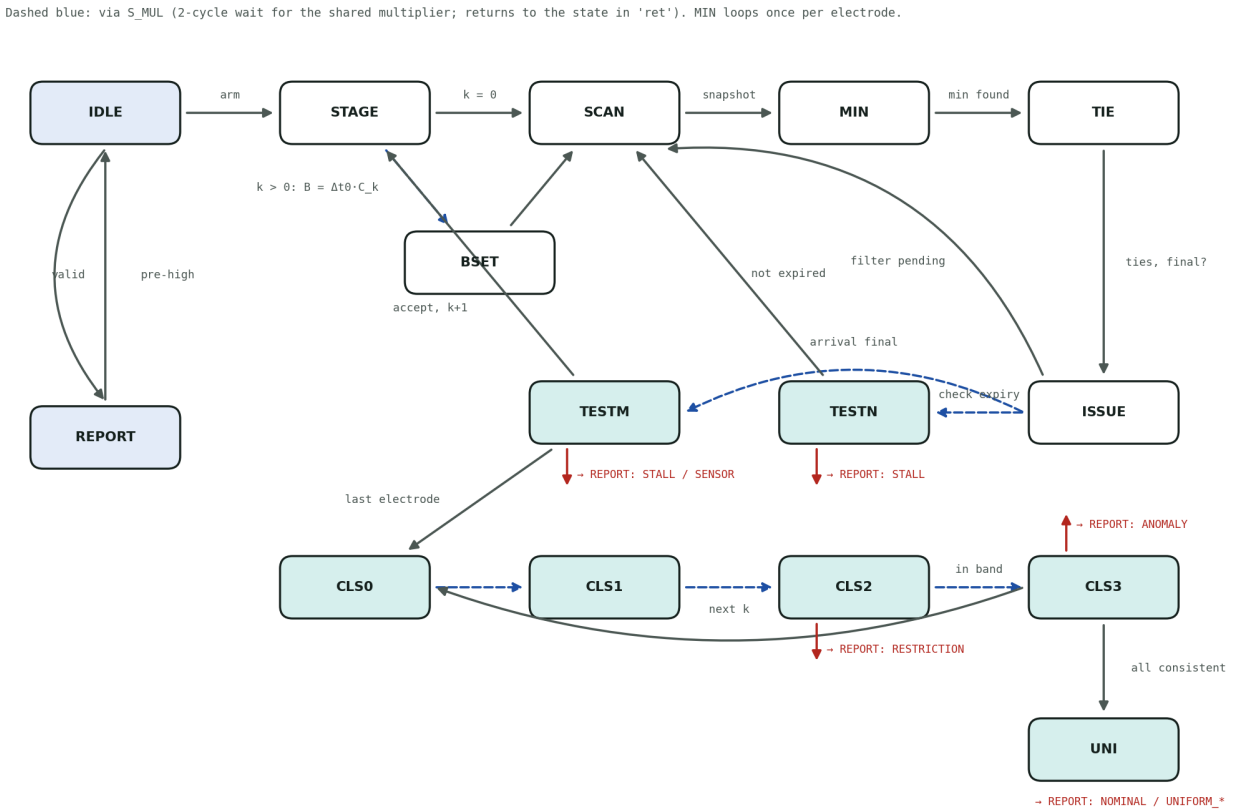


Figure 11.3. Sequencer FSM (16 states).

State	Role
IDLE	wait for arm; clear captures; pre-high check
STAGE	set the stage window bound B (constant or via multiplier)
BSET	store B from the multiplier
SCAN	snapshot all timestamps, flags and now
MIN	earliest stamp, one electrode per clock
TIE	tie mask; finality (now > m)
ISSUE	restart if a filter is pending; else issue the window test
TESTM	test the earliest arrival: STALL, SENSOR or accept
TESTN	test expiry at the sampled now: STALL or keep scanning
CLS0–CLS3	per-segment consistency tests (3 multiplies each)
UNI	uniform band on the total time
REPORT	one-cycle valid pulse with cls/loc
MUL	2-cycle wait for the shared multiplier, returns to 'ret'

11.4 Timing closure

The first implementation missed 100 MHz by 7.97 ns. The critical path was the combinational scan: a serial minimum over four timestamps, a compare and a subtract, 36 logic levels. Because latency does not affect results (§11.2), the scan was turned into a multi-clock walk over a registered snapshot. The worst path is now 5 logic levels, with +3.72 ns of slack.

11.5 Implementation results (Vivado 2022.2, xc7z020clg400-1, out-of-context)

Resource	Used	Available	Util %
Slice LUTs	535	53200	1.01
Slice Registers	674	106400	0.63
DSPs	3	220	1.36
Block RAM Tile	0	140	0.00

Clock 100 MHz: WNS = 3.720 ns -> Fmax ~ 159 MHz (post-route, out-of-context)

Power (vectorless estimate): total 0.046 W, dynamic 0.006 W

12 Verification

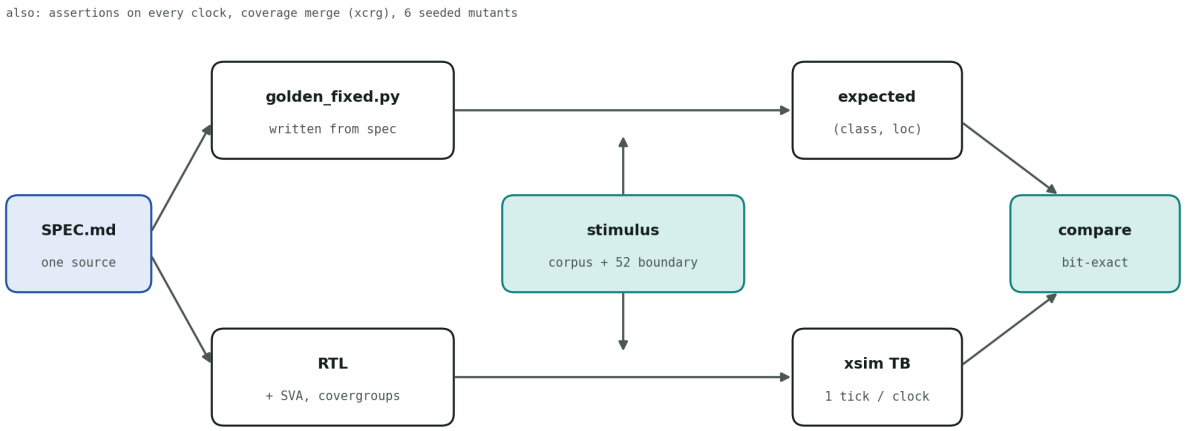


Figure 12.1. Verification flow.

12.1 Strategy

- **Independent reference.** `golden_fixed.py` is written from SPEC.md, not from the RTL; `golden_float.py` expresses the same rules as physical ratios. The two agree on 100 % of the corpus.
- **Testbench.** Plain SystemVerilog, file-driven, under Vivado xsim. It drives the tick itself and mirrors the DUT's count without hierarchical references. It ticks every clock, except for 3 clocks after an electrode rises (the synchronizer and capture latency), and it can inject glitch pulses of a given width.
- **Regression.** The corpus is split across parallel xsim processes (capped at SIM_JOBS = 4 after a 12-process run exhausted the machine's memory).

12.2 Results

Check	Result
Full corpus, RTL vs fixed-point reference	9,900 / 9,900 bit-exact; 1.68×10^{9} ticks simulated
Boundary suite (52 directed cases)	52 / 52 bit-exact
Assertions (final set) on the boundary suite	0 failures
Functional coverage	16/16 states, 30/30 transitions, all boundary events, 17/17 class × location
Mutation test (6 seeded RTL bugs)	6 / 6 caught by assertions (after two checks were strengthened)
Counter saturation	unreachable: worst undecided run 2,123,154 ticks of 16,777,215

12.3 Assertions

Assertion	Property
a_legal	only legal FSM transitions (explicit successor table)
a_mul_ret, a_mul_hold	multiplier wait returns to 'ret'; operands stable while waiting
a_idx_step, a_idx_range	stage index increases by at most 1 and stays in range
a_valid_src, a_valid_one, a_valid_run, a_arm_clean, final	exactly one result per run
a_now_step, a_now_nosat, a_tprev_mono	timebase advances by 0 or 1; never saturates; stage time monotonic

Assertion	Property
a_filter, a_ts_hold, a_ts_past, a_pend_x	glitch-filter rule; timestamps stable, in the past; pending excludes captured
a_arrived, a_order, a_dt, a_inwin, a_minfirst	every accepted stage arrived, in order, inside its window, plausible
a_stall_idx, a_stall_late, a_stall_expired	a STALL is only reported when its window has truly expired
a_loc_cls, a_loc_range	location consistent with class and within range

Window predicates in the checks are recomputed from the raw tolerances, not the folded ROM constants the RTL uses.

12.4 Mutation test

Mutant	Change	Wrong results	Assertion errors	Fired
classify_early	classify after N−1 electrodes	18	82	a_arrived, a_dt, a_inwin, a_order
no_order_check	accept out-of-order arrivals	6	16	a_arrived, a_dt, a_inwin, a_order
no_late_stall	accept arrivals after the window	3	3	a_inwin
no_min_first	drop the first-segment floor	3	3	a_minfirst
stall_early	stall on the window edge (≥ for >)	1	1	a_stall_late
short_filter	glitch filter one clock short	3	107	a_filter

Measured in the Phase 5 mutation run. `run.py phase5-mutants` reproduces it, but its result table has not been regenerated in the repository.

12.5 Boundary suite

Case	Arrival ticks (−1 never, −2 high before arm)	Glitches (e, tick, clocks)	Expected
nominal	42463, 84926, 127389, 169852	—	NOMINAL 0
win0_edge_in	212315, 424630, 636945, 849260	—	UNIFORM_SLOW 0
win0_edge_out	212316, -1, -1, -1	—	STALL 0
win1_edge_in	42463, 169852, -1, -1	—	STALL 2
win1_edge_out	42463, 169853, -1, -1	—	STALL 1
win2_edge_in	42463, 84926, 212315, -1	—	STALL 3
win2_edge_out	42463, 84926, 212316, -1	—	STALL 2
win3_edge_in	42463, 84926, 127389, 254778	—	RESTRICTION 3
win3_edge_out	42463, 84926, 127389, 254779	—	STALL 3
tol_hi_edge_in	42496, 93292, 135788, 178284	—	NOMINAL 0
tol_hi_edge_out	42496, 93293, 135789, 178285	—	RESTRICTION 1
tol_lo_edge_in	42496, 76692, 119188, 161684	—	NOMINAL 0
tol_lo_edge_out	42496, 76691, 119187, 161683	—	ANOMALY 1
minfirst_edge_in	21232, 42464, 63696, 84928	—	UNIFORM_FAST 0
minfirst_edge_out	21231, 42462, 63693, 84924	—	SENSOR 0
earliest_tick	1, 84926, 127389, 169852	—	SENSOR 0
uni_hi_in	48766, 97532, 146298, 195064	—	NOMINAL 0
uni_hi_out	48766, 97532, 146298, 195065	—	UNIFORM_SLOW 0
uni_lo_in	36160, 72320, 108480, 144640	—	NOMINAL 0
uni_lo_out	36159, 72318, 108477, 144639	—	UNIFORM_FAST 0
b2b_consecutive	42463, 42464, 42465, 42466	—	ANOMALY 1
b2b_last_pair	42463, 84926, 127389, 127390	—	ANOMALY 3

Case	Arrival ticks (−1 never, −2 high before arm)	Glitches (e, tick, clocks)	Expected
tie_e0_e1	42463, 42463, 127389, 169852	—	SENSOR 1
tie_mid	42463, 84926, 84926, 169852	—	SENSOR 2
all_same_tick	42463, 42463, 42463, 42463	—	SENSOR 1
out_of_order	42463, 127389, 84926, 169852	—	SENSOR 1
prehigh_e0	-2, 84926, 127389, 169852	—	SENSOR 0
prehigh_e1	42463, -2, 127389, 169852	—	SENSOR 1
prehigh_e2	42463, 84926, -2, 169852	—	SENSOR 2
prehigh_e3	42463, 84926, 127389, -2	—	SENSOR 3
prehigh_e1_e3	42463, -2, 127389, -2	—	SENSOR 1
prehigh_all	-2, -2, -2, -2	—	SENSOR 0
restr_seg1	42463, 106157, 148620, 191083	—	RESTRICTION 1
anom_seg1	42463, 67940, 110403, 152866	—	ANOMALY 1
restr_seg2	42463, 84926, 148620, 191083	—	RESTRICTION 2
anom_seg2	42463, 84926, 110403, 152866	—	ANOMALY 2
restr_seg3	42463, 84926, 127389, 191083	—	RESTRICTION 3
anom_seg3	42463, 84926, 127389, 152866	—	ANOMALY 3
missed_from_e0	-1, -1, -1, -1	—	STALL 0
missed_from_e1	42463, -1, -1, -1	—	STALL 1
missed_from_e2	42463, 84926, -1, -1	—	STALL 2
missed_from_e3	42463, 84926, 127389, -1	—	STALL 3
missed_all	-1, -1, -1, -1	—	STALL 0
missed_last	42463, 84926, 127389, -1	—	STALL 3
missed_middle	42463, -1, 127389, 169852	—	SENSOR 1
glitch_e0_w1	42463, 84926, 127389, 169852	(0,1000,1)	NOMINAL 0
glitch_e2_stage1_w1	42463, 84926, 127389, 169852	(2,63694,1)	NOMINAL 0
glitch_e0_w3	42463, 84926, 127389, 169852	(0,1000,3)	NOMINAL 0
glitch_e2_stage1_w3	42463, 84926, 127389, 169852	(2,63694,3)	NOMINAL 0
glitch_e0_w4	42463, 84926, 127389, 169852	(0,1000,4)	SENSOR 0
glitch_e2_stage1_w4	42463, 84926, 127389, 169852	(2,63694,4)	SENSOR 1
glitch_all_lines_w3	42463, 84926, 127389, 169852	(0,500,3), (1,500,3), (2,500,3), (3,500,3)	NOMINAL 0

12.6 Not verified

- No formal proof (no formal tool installed); mutation testing was used instead.
- A single simulator (xsim). iverilog lacks covergroups and concurrent assertions.
- The full corpus was not rerun with the final assertion set; Phase 4's bit-exact run used the same RTL.
- No hardware run on the PYNQ-Z2 board (Phase 7 not done).

13 What we claim, and what we do not

13.1 Claims supported by evidence

Claim	Evidence
Multi-point arrival timing detects filling faults better than a fixed timeout or a single flow-time check at equal false-alarm rate	88.9 % vs 46.8 % / 55.9 % (test split, §9)
The detector classifies physically distinguishable faults and locates restrictions and stalls	stall 100 % right gap; strong restriction 90 % right segment; confusion matrix §9
Some causes are provably indistinguishable from arrival times	Proposition 1, identical ticks (§6)
Restriction visibility has a computable threshold	10/13/15 mm by segment (§6.4)
The detector fits in a small FPGA block at 100 MHz	535 LUT, 3 DSP, WNS +3.72 ns (§11)
The hardware implements the specification exactly	9,900/9,900 bit-exact, 52 boundary cases, assertions, coverage, 6/6 mutants (§12)
The base paper's flow-time algorithm is reproduced	3/3 flow times inside the paper's IQRs (§10)

13.2 Where it is not superior

- **Detection versus a fixed-window ablation:** no better (89.4 % vs 88.9 %). The gain over the ablation is classification, not detection.
- **False alarms:** 6–13 % at realistic jitter, as high as the baselines and too high for a clinical device.
- **Latency:** slower than the fixed-window variant for viscous samples (342 s vs 87 s median).
- **Evidence base:** weaker than the base paper, which used real devices and clinical samples.

13.3 Not covered (limitations)

- The four-electrode results are simulated; no lateral strip was built or measured. The generator shares the detector's physics family, so performance is an upper bound.
- The calibration uses one membrane, water, a vertical-rise figure read by eye, with a +9 % gravity systematic and an early-time deviation from Washburn.
- The real-data check is one run of a single-electrode device; it validates timing only.
- Mild restrictions (ΔL_{eq} below 10–15 mm) are not detected; weak upstream restrictions look like slow fluids; a sensor fault's electrode index is ambiguous; a missed last electrode looks exactly like a stall.
- Assumed values: sample volume 25 µL, creep τ 20 s, electrode positions, textbook plasma/blood viscosities. The 1.8 s lag is applied per arrival, which is our interpretation of the paper's flow-time bias.
- Not modelled: evaporation, temperature, swelling, partial saturation behind the front.
- Hardware: out-of-context timing, vectorless power, no board run; the interface assumes off-chip comparators; no sampled-signal front end ('interface B'); the 10 µs glitch filter length was chosen, not derived from measured noise.
- The immunoassay quantification of the base paper is out of scope.

13.4 Threats to validity

- **Circularity:** the detector's physics is a simplification of the generator's. Mitigated by the richer generator (series resistance, finite volume, creep, sensor faults), but not removed.

- **Tuning:** thresholds were chosen on calibration seeds and reported on test seeds from the same generator; a real deployment would need re-tuning on measured strips.
- **Statistics:** the false-alarm rate rests on 750 healthy test strips (95 % CI 5.4%–9.1%).

14 Project log: what was done, step by step

Phase	What was done	Outcome
0 Plan	Environment survey; architecture options (fixed windows, segment ratios with online K, sampled front end); PLAN.md	approved
Proposal	One-page A4 PDF generated with ReportLab	group number and names still placeholders
1 Tracer	generator → reference model → RTL in xsim → compare, 3 scenarios	pass; first localization-lag finding
2 Physics	literature search; digitization; K fit; Darcy generator; sensor model; scaling	K = 9.42 mm²/s; 7/7 checks
3 Golden	9,900-run corpus; grid-search tuning; baselines; metrics; SPEC v0.2	detection 88.9 % vs 46.8 %
4 RTL	timestamp-capture redesign; sharded regression; Vivado OOC P&R; timing fix	9,900/9,900; WNS +3.72 ns
5 Verification	SVA, coverage, 52 boundary cases, mutation test, parallelism cap	all pass; 6/6 mutants
6 Real data	Bezinge files; Python port; analytic self-test; comparison with the paper	3/3 inside IQR; lag and jitter confirmed
8 Report	identifiability computations; report draft; HTML docs; this document	done
7 PYNQ demo	optional board demo	not started

14.1 Problems met and how they were solved

Problem	Fix
cmd.exe splits arguments on '=' and the project path has spaces, breaking xsim plusargs	fixed file names in each simulation directory
xvlog's own log file collided with the console redirect	console output to *.out files
Vivado read_xdc and xcrg split paths at spaces	Tcl list / relative paths
Corpus needed ~10^{10} simulated cycles	timestamp-only RTL; testbench ticks every clock; parallel shards
Post-route timing missed by 7.97 ns	multi-clock snapshot scan
Two seeded bugs escaped the first assertions	generalized accepted-stage checks; added a glitch-filter assertion
12 parallel simulators with checks crashed the PC	SIM_JOBS cap (default 4); phase 5 uses one process
xsim mis-evaluated sized casts in coverage bin ranges (33.6 % reported)	true coverage recovered from hit values; bins rewritten with integers
Publisher pages bot-checked; ETH rate-limited	open MDPI paper for calibration; user-supplied dataset

14.2 Decisions that differ from the original handoff

Topic	Handoff	Decision and reason
Calibration source	Mendez 2010 / Fu 2011	Kasetsirikul 2020: open, lateral-flow membrane
Upstream restriction	never distinguishable	distinguishable above ~15 mm (Proposition 3)
Uniform test	first segment	total fill time (lag matters less)
Formal check	optional	no formal tool; mutation testing instead
Octave reproduction	run plotResults.m	not installed; Python port with an analytic self-test
Interface B, PYNQ demo	optional	not done: single-electrode data; time budget

15 Reproduction guide

Environment: Windows, Python 3.12 (run as py -3.12), NumPy, SciPy, Matplotlib, ReportLab, Vivado 2022.2 (xsim, xcrg, synthesis). No simulator is needed for phases 2, 3, 6 and 8 or for this document.

```
py -3.12 run.py phase1          # tracer: 3 scenarios through xsim
py -3.12 run.py phase2          # calibration, scaling, generator checks
py -3.12 run.py phase3          # corpus, tuning, baselines, metrics (Python only)
py -3.12 run.py phase4          # full RTL regression (SIM_JOBS parallel xsim) + Vivado
py -3.12 run.py phase5          # boundary suite, assertions, coverage (one xsim process)
py -3.12 run.py phase5-mutants  # mutation test (SIM_JOBS at a time)
py -3.12 run.py phase6          # base-paper data reproduction (Python only)
py -3.12 run.py phase8          # identifiability assets (Python only)
py -3.12 report/build_master_pdf.py   # this document
```

Path	Contents
PLAN.md, SPEC.md	plan with status log; detector specification
physics/	parameters, calibration, Darcy generator, sensor, scaling
scenarios/	corpus, boundary cases, tracer
golden/	reference models, baselines, evaluation, identifiability
rtl/, tb/, syn/	design, testbench + checks, Vivado script
realdata/, data/	base-paper port; literature and real data with provenance
results/	generated figures and tables
report/	draft report, this document and its builder
docs/	HTML documentation and study guide
proposal/	one-page proposal

Appendix A Detector specification (SPEC.md)

```
# Detector specification (v0.2, Phase 3)

Changes from v0.1, both driven by Phase 3 corpus results:
- **First-segment plausibility floor.** No plausible sample wicks more than 2× faster than nominal: water at 37 °C is 1.29×. A first segment faster than MIN_FIRST_Q/256 of E_0 is therefore a sensor fault, not a fluid. Under v0.1 a spurious early trigger on electrode 0 shrank r̂ and caused false STALLs.
- **Uniform-K test on the total fill time.** The test now uses t_{N−1} = ΣΔt against E_TOT = ΣE instead of Δt_0. The constant detection lag and the jitter are then a small fraction of the measured interval. Under v0.1, faster fluids hid inside the band.

This is the single source the golden models and the RTL are written from. Neither is translated from the other.

## Inputs and timebase
- N electrodes at distances L_0 < … < L_{N-1} from the inlet. The inlet is at L = 0.
- Tick = 1 ms. `arm` marks sample application (t = 0). The timestamp counter `now` (24 bit) counts ticks from arm.
- The arrival of electrode k is the tick t_k ≥ 1 at which its line is first seen high. A line already high at arm is *pre-high*.

## Constants (generated from `physics/params.py`, integer)
- E_k = round((L_k² − L_{k−1}²)/K_nom · 1000), with L_{−1} = 0. This is the expected segment time in ticks.
- E_TOT = ΣE_k, the expected total fill time.
- Tolerances are in units of 1/256: TOL (segment consistency), TOL_ABS (nominal band), TOL_STALL (stall margin), FIRST_Q (maximum first-segment time, as a multiple of E_0) and MIN_FIRST_Q (minimum first-segment time).

## Stage logic (stage idx = waiting for electrode idx; t_prev = t_{idx−1}, with t_{−1} = 0)
Segment times: Δt_k = t_k − t_{k−1}.

The **timeout predicate** at elapsed e = now − t_prev is:
- idx = 0: e·256 > E_0·FIRST_Q
- idx ≥ 1: e·E_0·256 > Δt_0·E_idx·(256 + TOL_STALL)

This is the adaptive window: the expected segment time is scaled by the measured first-segment ratio r̂ = Δt_0/E_0.

Rules, in priority order:
0. At arm, any pre-high line → **SENSOR**, loc = lowest pre-high index.
1. If the timeout predicate becomes true before any not-yet-captured electrode j ≥ idx arrives, the result is **STALL** with loc = idx. An arrival whose own elapsed time already satisfies the predicate is too late, so the timeout wins ties.
2. Otherwise let te be the earliest arrival tick among the electrodes j ≥ idx, and A the set of electrodes arriving at te. If A contains any j ≠ idx, the result is **SENSOR** with loc = (min of A\{idx}) if idx  in  A, else idx.
3. Otherwise capture Δt_idx = te − t_prev. If idx = 0 and Δt_0·256 < E_0·MIN_FIRST_Q, the result is **SENSOR** with loc = 0 (implausibly fast first segment). Otherwise move to the next stage. After the last electrode, classify.

## Classification (all comparisons are exact integer arithmetic; no division and no rounding)
For k = 1 … N−1, in order, the first k satisfying either test gives the result:
- Δt_k·E_0·256 > Δt_0·E_k·(256 + TOL) → **RESTRICTION**, loc = k (extra resistance first appears in segment k)
- Δt_k·E_0·256 < Δt_0·E_k·(256 − TOL) → **ANOMALY**, loc = k (a downstream segment is faster than the physics predicts)

If no k triggers, all segments are consistent with one K. Test it on the total fill time T = t_{N−1}:
- T·256 > E_TOT·(256 + TOL_ABS) → **UNIFORM_SLOW**
- T·256 < E_TOT·(256 − TOL_ABS) → **UNIFORM_FAST**
- otherwise → **NOMINAL**

For these three classes loc = 0.

## Hardware realisation (rtl/, Phase 4)
- **Arrival tick.** t_k is the value of `now` in the clock where electrode k's synchronized line starts a high run that lasts DEB clocks (the glitch filter). Filtering adds latency, never timestamp error.
- **Order-independence.** The sequencer acts only on captured timestamps. It treats an arrival as final when no filter is pending and `now` > t; it treats a timeout as final when the window predicate holds at a sampled `now`, because the predicate is monotonic. So the outcome is the SPEC function of the ticks and does not depend on the clock-to-tick ratio or the multiplier latency. The regression exploits this: it runs one tick per clock.
- **Arithmetic.** One shared 24 × 36-bit multiplier computes (measured time) × (ROM constant). params_pkg folds every constant × constant product. There is no divider.

## Class codes (3 bit)
NOMINAL 0 · UNIFORM_SLOW 1 · UNIFORM_FAST 2 · RESTRICTION 3 · STALL 4 · SENSOR 5 · ANOMALY 6

## Identifiability notes
- UNIFORM_* cannot say which fluid property changed.
- A restriction in segment 0 (upstream of electrode 0) adds a constant resistance R0, so t ~ L² + 2·R0·L. Its segment
```

ratios *decrease* downstream (Phase 2 generator: ρ = 2.2, 1.8, 1.6, 1.5 for k/6 over 4 mm), and it is flagged as ANOMALY. Only a weak upstream restriction (2·R0·L << L², i.e. deviation within TOL) is indistinguishable from UNIFORM_SLOW. This refines the handoff's limitation; Phase 3 quantifies the boundary.

Appendix B References

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[7] H. Bruus, Theoretical Microfluidics, Oxford University Press (2008).